KIAN ENTEZARI

Computer Vision | Robotics | Embedded Systems

(925) 808-1559 | kianxyz@gmail.com | Portfolio | GitHub | LinkedIn

EDUCATION

University of California, Berkeley *Expected Spring 2028*

B.S., Electrical and Computer Engineering | GPA: 4.00

TECHNICAL SKILLS

Languages: Python, C, C++, C#
Perception: DIAVision, OpenCV, YOLOv8, LiDAR
Machine Learning: PyTorch, scikit-learn, ONNX Runtime
Neurotech: MNE-Python, pyRiemann, EEGNet
Tools: AutoCAD, Git, ROS 2, MuJoCo, Isaac Sim
Protocols: Modbus TCP, Google API, Meta API, Open AI API
Electronics: KiCad, LTspice, PCB design
Embedded Platforms: ESP32, Arduino

EXPERIENCE

AI & Robotics Researcher | Berkeley Artificial Intelligence Research (BAIR) - Speech Group *Oct 2026 - Present*

- Design and develop VERA, an AI-powered humanoid robotic head with expressive eye, neck, and jaw mechanisms for natural human-robot interaction.
- Simulate and control eye, neck, and jaw mechanisms using MuJoCo and Isaac Sim, modeling joint dynamics and actuator control.
- Implement sim-to-real control pipelines that translate AI-generated facial expressions into synchronized physical movements.

Computer Vision Engineer Intern | Mantis Robotics *Nov 2025 - Aug 2026*

- Developed a stereo vision pipeline for the MR-1 robotic arm with hand-eye calibration for camera-to-robot coordinate transformation, achieving multi-object 3D localization with <1 mm accuracy while cutting hardware costs by over 90% by replacing industrial cameras with two low-cost webcams.
- Integrated YOLOv8 object detection (OpenCV, ONNX Runtime) for real-time object detection and class identification.
- Built a LiDAR-camera fusion pipeline using YOLO detections and projected point clouds for 3D object localization.
- Created a Modbus TCP server to transmit robot coordinates from the vision pipeline to the robot controller, and integrated digital I/O for robotic hardware communication and control.

Software Engineer Intern | Inner Health & Wellness *Oct 2025 - Jan 2026*

- Built and deployed an AI health and wellness chatbot using the OpenAI API.
- Developed backend services integrating Google and Meta APIs for content ingestion and in-app messaging, deployed on Vercel.

PROJECTS

Neural Relay | Brain Computer Interface *Nov 2026*

- Designed and built a brain-computer interface that enables one person to control another person's hand movements through decoded motor imagery signals.
- Developed Raspberry Pi firmware for real-time 8-channel EEG signal acquisition, processing, and communication with the control system.
- Implemented a motor imagery decoding pipeline using Riemannian tangent-space features, logistic regression, and CNNs, achieving 74.6% balanced accuracy on EEG recordings with logistic regression.

Heart Beat Classifier | ECG Signal Classification using Machine Learning *Nov 2026*

- Developed a 1D CNN for atrial fibrillation detection from 2,000-sample ECG recordings, benchmarking against a feature-engineered Random Forest achieving a best validation AUROC of 0.85.

Independent Research | Multi-Analyte Electrochemical Biosensor *May 2026 - Aug 2026*

- Designed a low-cost TLV272-based potentiostat for electrochemical detection of glucose, lactate, cortisol, alcohol, and vitamin C.
- Developed a reverse iontophoresis system for non-invasive interstitial fluid extraction and continuous biosensing.